

Master 2 Informatique
Data sciences, infrastructure cloud et sécurité

Machine Learning By Practice

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General Overview

- Introduction
- Data Visualization
- Dimensionality Reduction
- **Supervised Learning**
- Unsupervised Learning
- Reinforcement Learning
- Conclusion

CM #4

Supervised Learning Learning with Trees

Learning with Trees

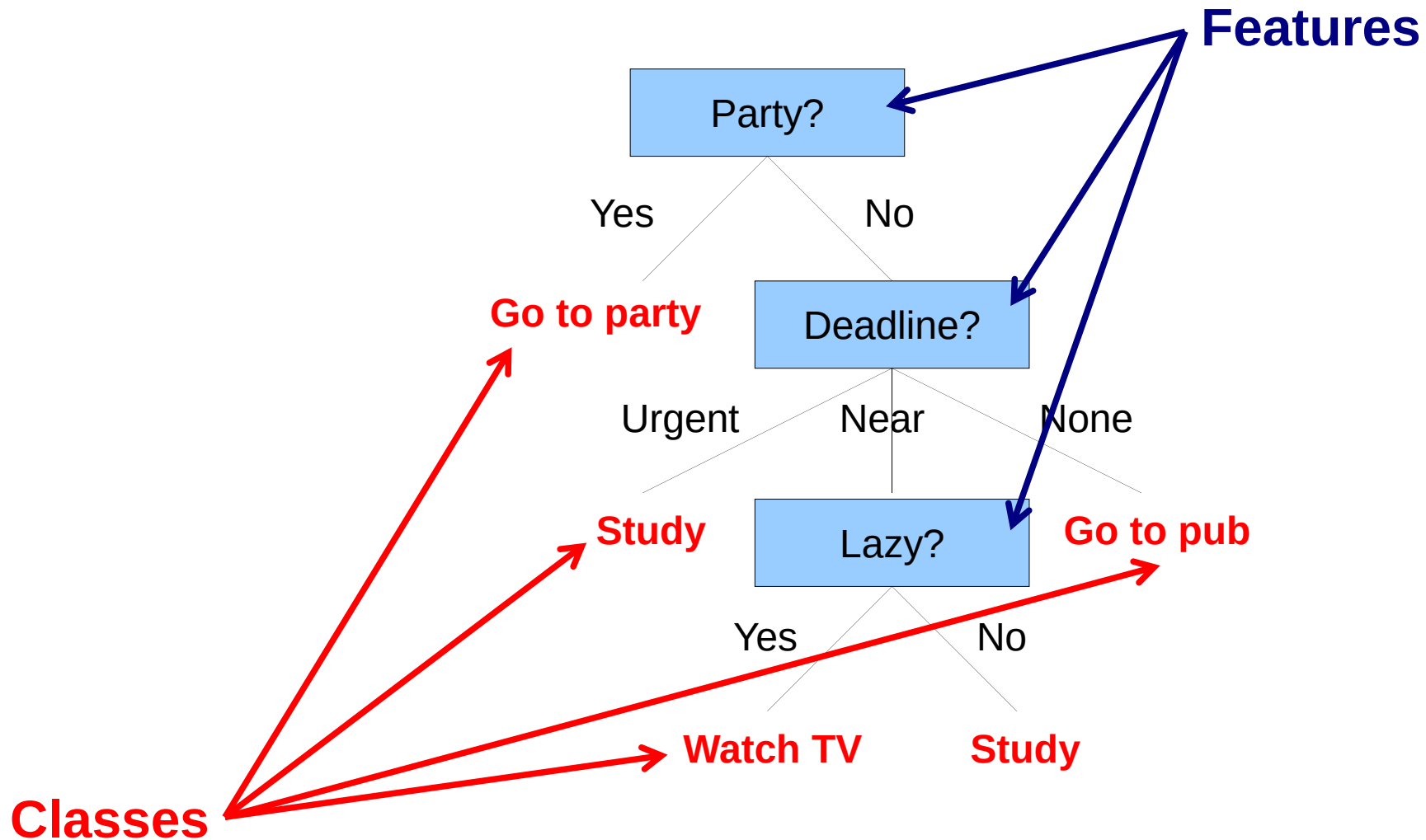
Supervised Learning Using Decision Trees

Decision trees are easy to understand.
They can be turned into a set of if-then rules.

What to do in the evening?



Supervised Learning Using Decision Trees



{Yes, No, Urgent, Near, None} are the feature values.

Supervised Learning

Building Decision Trees

How to build decision trees from examples?

Urgent	Yes	Yes	Party
Urgent	No	Yes	Study
Near	Yes	Yes	Party
None	Yes	No	Party
None	No	Yes	Pub
None	Yes	No	Party
Near	No	No	Study
Near	No	Yes	TV
Near	Yes	Yes	Party
Urgent	No	No	Study

Learning with Trees

Building Decision Trees

The computational cost of making the tree is fairly low ($O(N \log N)$ for binary trees), and the cost of using it is even lower ($O(\log N)$ for balanced binary trees) where N is the number of datapoints.

All decision tree algorithms build the tree in a greedy manner starting at the root node, choosing the most informative feature at each stage.

Several algorithms:

- **ID3** for classification
- C4.5 (extension of ID3) for classification
- CART builds binary trees for classification and regression

Learning with Trees

Building Decision Trees

The entropy of a set of probabilities p_i is:

$$Entropy(p) = - \sum_i p_i \log_2 p_i$$

Suppose we have a set of positive and negative examples with one feature that can only take on two values...

If all examples are positive, we don't get any extra information from knowing the value of the feature.

However if the feature separates the examples into 50% positive and 50% negative, then the entropy is at a maximum and knowing about that feature is very useful to us.

Learning with Trees

Building Decision Trees

At each stage, ID3 selects the feature that decreases the entropy of the whole training set the least.

ID3 computes the information gain for each feature and selects the one that has the highest value.

The information gain of a feature F for a set of examples S is given by:

$$Gain(S, F) = Entropy(S) - \sum_{f \in values(F)} \frac{|S_f|}{|S|} Entropy(S_f)$$

where :

- S_f is the set of examples in S that have value f for feature F .
- $|S_f|$ is the number of examples in S that have value f for feature F .

Learning with Trees

ID3 Algorithm

- If all examples have the same label:
Return a leaf with that label
- Else if there are no features left:
Return a leaf with the most common label
- Else:
 - Choose the feature F that maximises the information gain of S to be the next node
 - Add a branch from the node for each possible value f in F
 - For each branch:
 - Compute S_{new} by removing F from the set of features
 - Recursively call the algorithm with S_{new}